

Independent CPT and F/O MCDUs

Each MCDU has its own page, scratchpad and messages, on the one shared FMS

Aircraft	A32NX
Where to find it	The two MCDUs on the pedestal
PR branch	pr/08-a32nx-mcd�-independent (independent; the close-up cockpit model needs a companion change in aircraft-large-files)
Status	Tested in the sim (September 2026).
Pictures	The MCDU gauge in the local browser harness, running the build installed in the sim; the gauge draws the CPT screen in its left half and the F/O screen in its right half, shown here side by side.

1. What it is for

2. What is per MCDU and what is shared

3. Messages
4. Power

5. Limitations and why

1. What it is for

The two MCDUs mirrored each other: same page, same scratchpad. On the aircraft each MCDU is its own display, working on the shared FMGC data (A320 FCOM DSC-22_10-10). Now each pilot can work on a different page.

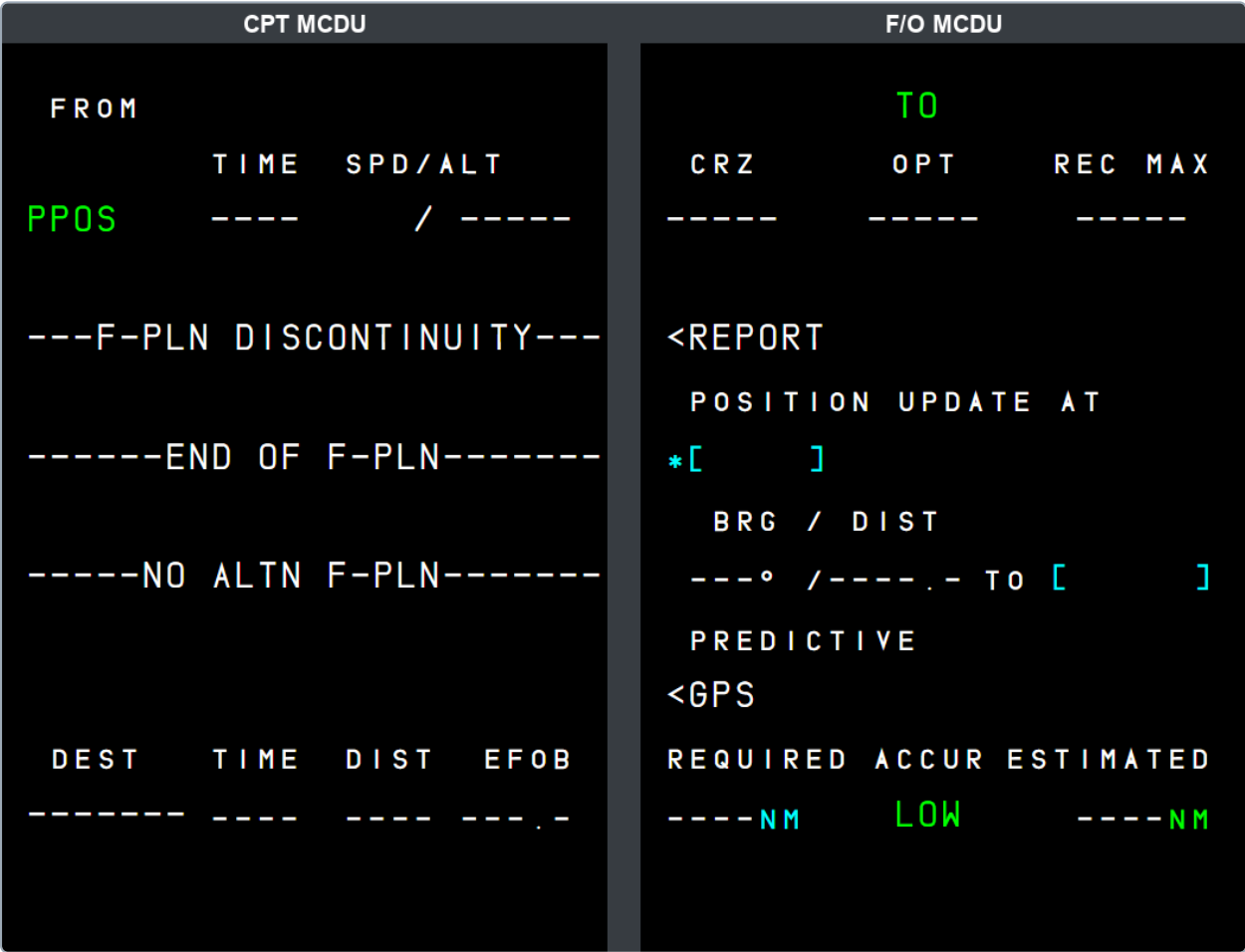


Figure 1. CPT on F-PLN, F/O on PROG at the same time.

2. What is per MCDU and what is shared

Per MCDU	Shared (one FMS)
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The page shown, the line keys	The flight plans, performance, fuel, navigation data
The scratchpad and what is typed in it	The data entered from either MCDU
The scratchpad messages (FORMAT ERROR, NOT ALLOWED ...) of an entry	FMGC messages (queued on both)
The keyboard entry (the MCDU that was clicked)	The EFIS symbols, the guidance

CPT MCDU	F/O MCDU
FROM	TO
TIME SPD/ALT	CRZ OPT REC MAX
PPOS --- / ---	-----
---F-PLN DISCONTINUITY---	<REPORT
-----END OF F-PLN-----	POSITION UPDATE AT
-----NO ALTN F-PLN-----	*[]
	BRG / DIST
	---° /-----.- TO []
	PREDICTIVE
	<GPS
DEST TIME DIST EFOB	REQUIRED ACCUR ESTIMATED
-----	---NM LOW ---NM
LFBO	250/12

Figure 2. Each scratchpad keeps its own text: LFBO on the CPT, 250/12 on the F/O.

3. Messages

CPT MCDU	F/O MCDU
FROM	INIT ↔
TIME SPD/ALT	CO RTE FROM/TO
PPOS --- / ---	----- REQUEST*
---F-PLN DISCONTINUITY---	ALTN/CO RTE INIT
-----END OF F-PLN-----	-----/-----
-----NO ALTN F-PLN-----	FLT NBR

	WIND/TEMP>
	COST INDEX TROPO
	--- 36090
DEST TIME DIST EFOB	CRZ FL/TEMP GND TEMP
-----	----- /---° ---°
LFBO	FORMAT ERROR

Figure 3. A wrong entry on the F/O MCDU: FORMAT ERROR there only.

CPT MCDU	F/O MCDU
FROM	INIT ↔
TIME SPD/ALT	CO RTE FROM/TO
PPOS --- / ---	----- REQUEST*
---F-PLN DISCONTINUITY---	ALTN/CO RTE INIT
-----END OF F-PLN-----	-----/-----
-----NO ALTN F-PLN-----	FLT NBR

	WIND/TEMP>
	COST INDEX TROPO
	--- 36090
DEST TIME DIST EFOB	CRZ FL/TEMP GND TEMP
-----	----- /---° ---°
LFBO	XYZ

Figure 4. CLR on the F/O MCDU: its message clears and its entry comes back; the CPT scratchpad is untouched.

Situation	What you should see
An entry is rejected	The message shows on the MCDU where the entry was made.
An FMGC message (e.g. CHECK TAKE OFF DATA)	Queued on both MCDUs; CLR on one clears it there only.
The FMS changes pages by itself (flight phase change on PROG / PERF / IDENT, engine-out page, INIT B to FUEL PRED)	Applies to both MCDUs.
ATSU system status (not the answer to an entry)	On the ATSU scratchpad of both MCDUs.
Typing with the PC keyboard	Goes to the MCDU that was clicked last; only that MCDU animates its keys.
SimBridge remote MCDU	CPT screen on its left side, F/O screen on its right side.

4. Power

- CPT MCDU: AC ESS SHED bus. F/O MCDU: AC 2 bus.
- An unpowered MCDU ignores its keys; the other one keeps working.

5. Limitations and why

Limitation	Why	What to do
No return to the MENU page after a power loss.	Same as the CPT MCDU before this change.	-
Close-up cockpit model: both screens show the whole texture until the companion change is merged.	The 4 UV values of the high-detail model (LOD00) live in the aircraft-large-files repository; this PR only changes LOD01.	Merge the aircraft-large-files PR first.